INCH-POUND
AA-C-101
July 5, 2017

FEDERAL SPECIFICATION

# CABINETS, HAZARDOUS MATERIAL (HM), SHIPBOARD QUALIFIED

The General Services Administration has authorized the use of this federal specification by all federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification establishes the requirements for hazardous material (HM) cabinets (formerly known as lockers) that are qualified for shipboard use.

1.2 Classification. HM cabinets that are shock and vibration qualified are of the following types and classes, as specified (see 6.2 and 6.4):

1.2.1 Types.

a. Type I – Flammable cabinet, shock and vibration qualified, yellow
b. Type II – Corrosive cabinet, shock and vibration qualified, dark blue

1.2.2 Classes.

a. Class 1 – 12 gallon
b. Class 2 – 15 gallon
c. Class 3 – 30 gallon

## 2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

MILITARY SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MILITARY STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

(Copies of these documents are available online at http://quicksearch.dla.mil.)

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 7125

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 30 - Flammable and Combustible Liquids Code

(Private sector and civil agencies may purchase copies of this voluntary standard from the National Fire Protection Association, 1401 K St. NW #500, Washington, DC 20005 or online at www.nfpa.org.)

3. REQUIREMENTS

3.1 Qualification. The HM cabinets furnished under this specification shall be products qualified for listing on the applicable qualified products list (see 4.3 and 6.3).

3.2 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

3.3 Installation. Shipboard HM cabinets shall be positively fixed to a deck or a foundation built on a ship's deck. The cabinet bottom pad or the cabinet mounting feet shall be welded to the ship's deck or welded to a foundation that is welded to the deck.

3.4 Design and construction. Shipboard qualified HM cabinets are designed to store groups of HM based on the primary hazard (such as flammables or corrosives). In general, design requirements specify features such as size, color, mounting methods, closing mechanism, locking ability, containment sumps, and shock and vibration qualification as specified in 3.5 and 3.6.

3.5 Physical requirements. Refer to table I for a summary of requirements.

3.5.1 Cabinet color. Color requirements of shipboard qualified HM cabinets are as follows:

a. Type I (all classes) - yellow, per industry standard.

b. Type II (all classes) - dark blue, per industry standard.

3.5.2 Identification plate(s)/label(s). Shipboard qualified HM cabinets shall be provided with permanently attached plate(s) or label(s) affixed to the cabinet exterior containing the following data:

a. Manufacturer name

b. Type, class, and associated nomenclature of cabinet (i.e., type I, class 1, flammable cabinet, shock and vibration qualified)

c. Cabinet capacity, in gallons

d. Contractor and Government Entity (CAGE) number

e. Part or identification number (see 6.4)

3.5.3 Hazard warning label plate(s)/label(s). Shipboard qualified HM cabinets shall be provided with permanently attached plate(s) or label(s) affixed to the cabinet exterior in accordance with this section. The label(s) or plate(s) shall be fabricated in accordance with requirements of NFPA 30 and shall be provided with the following inscriptions and content label(s):

a. For type I flammable cabinet, shock and vibration qualified (all classes):

Inscription:
FLAMMABLE
KEEP FIRE AWAY



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b. For type II, corrosive cabinet, shock and vibration qualified (all classes):

Inscription:
HAZARDOUS
KEEP FIRE AWAY

Content Identification Label:



**CORROSIVE**
**STORAGE CABINET**

3.5.4 Cabinet shelves. Shipboard qualified HM cabinets shall be provided with a minimum of two shelves per cabinet. Cabinet shelves shall be height adjustable. Additional details regarding corrosive cabinet shelf requirements can be found in 3.6.8.

3.6 Performance requirements. Refer to table I for a summary of requirements.

3.6.1 Shock resistance. Type I and type II cabinets shall meet the grade B shock acceptance requirements of MIL-DTL-901 when tested in accordance with 4.5.1.

3.6.2 Vibration resistance. Type I and type II cabinets shall meet the type I vibration requirements of MIL-STD-167-1 when tested in accordance with 4.5.2.

3.6.3 Fire protection rating. Type I and type II cabinets shall meet the criteria specified in NFPA 30.

3.6.4 Spill containment. Type I and type II cabinets shall have secondary containment or a cabinet sump that contains a minimum of 10 percent of the aggregate cabinet volume when tested in accordance with 4.5.3.

NOTE:

This spill containment requirement may be met by providing a 2-inch raised sill designed to retain liquid as long as the 2-inch sill depth equates to a sump design that can hold as least 10 percent of the aggregate cabinet volume. The 2-inch raised sill is required by NFPA 30 for flammable liquid storage cabinets.

3.6.5 Self-closing, self-latching doors. Type I and type II cabinets shall have self-closing, self-latching doors when tested in accordance with 4.5.4.

3.6.6 Lockability. Type I and type II cabinets shall be lockable by padlock or integral lock when tested in accordance with 4.5.5.

3.6.7 Steel with acid-resistant epoxy paint. Type II cabinets shall be of steel construction with acid-resistant epoxy paint or epoxy powder coating when tested in accordance with 4.5.6.

3.6.8 Lined cabinet shelves. Type II cabinet shelves shall be either galvanized metal or lined with acid-resistant material (epoxy paint or epoxy powder coating). In addition, type II cabinet shelves shall be fitted with polyethylene trays or liners to contain spillage when tested in accordance with 4.5.7.

TABLE I. Summary of requirements.

| Requirement | Requirement paragraph | Applicability | |
|---|---|---|---|
| | | Type I | Type II |
| Color | 3.5.1 | Yellow | Dark blue |
| Identification plate | 3.5.2 | X | X |
| Hazard warning plate | 3.5.3 | X | X |
| Minimum of two height adjustable shelves | 3.5.4 | X | X |



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TABLE I. Summary of requirements – Continued.

| Requirement | Requirement paragraph | Applicability | |
|---|---|---|---|
| | | Type I | Type II |
| Shock resistance | 3.6.1 | X | X |
| Vibration resistance | 3.6.2 | X | X |
| Fire protection rating | 3.6.3 | X | X |
| Spill containment | 3.6.4 | X | X |
| Self-closing, self-latching doors | 3.6.5 | X | X |
| Lockable | 3.6.6 | X | X |
| Steel with acid-resistant epoxy paint | 3.6.7 | | X |
| Lined cabinet shelves | 3.6.8 | | X |

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of examinations and tests. The inspection requirements specified herein are classified as follows:

a. Qualification tests (see 4.3).

b. Quality conformance tests (see 4.4).

4.3 Qualification tests. Qualification tests shall consist of all of the tests listed in table II.

4.3.1 Qualification inspection. Qualification inspection shall include the examination and inspection of results of testing listed in table II.

4.3.2 Design changes. Any redesign or modification of the manufacturer's standard product not qualified in accordance with 4.3.1 shall constitute rejection of the cabinets.

4.4 Quality conformance inspection. Quality conformance inspection shall include specific tests, examinations, and inspections listed in table II.

4.4.1 Lot. All unmixed material of the same type offered for delivery at one time shall be considered a lot for purposes of quality conformance testing.

4.4.2 Sampling for quality conformance testing. A sufficiently large production sample of each cabinet shall be selected in order to permit the performance of all quality conformance tests.

4.4.3 Small lots. Quality conformance tests shall not be required on any delivery of less than 100 cabinets. However, deliveries of such small quantities that have not been tested shall be so identified in the manufacturer's records. Once the accumulated total of such untested deliveries reaches 100 cabinets, sample material shall be selected and subjected to the required quality conformance tests identified in table II.

4.4.4 Inspection conditions. Specific inspection conditions shall not be required. Samples may be inspected under normal factory or warehouse conditions.



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TABLE II. Tests, examinations, and inspections.

| Test requirement | Requirement paragraph | Test method paragraph | Qualification testing required | Quality conformance testing required | Applicability (Type[s]) |
|---|---|---|---|---|---|
| Shock requirement | 3.6.1 | 4.5.1 | X | | I, II |
| Vibration resistance | 3.6.2 | 4.5.2 | X | | I, II |
| Spill containment | 3.6.4 | 4.5.3 | X | | I, II |
| Self-closing, self-latching doors | 3.6.5 | 4.5.4 | X | X | I, II |
| Lockable | 3.6.6 | 4.5.5 | X | X | I, II |
| Steel with acid-resistant epoxy paint | 3.6.7 | 4.5.6 | X | | II |
| Minimum of two height adjustable shelves | 3.5.4 | 4.5.7 | X | X | I, II |
| Lined cabinet shelves | 3.6.8 | 4.5.7 | X | X | II |
| Color, type I | 3.5.1 | 4.5.8 | X | X | I |
| Color, type II | 3.5.1 | 4.5.8 | X | X | II |
| Identification plate | 3.5.2 | 4.5.9 | X | X | I, II |
| Hazard warning plate | 3.5.3 | 4.5.10 | X | X | I, II |

4.5 Tests.

4.5.1 Shock qualification. Type I and type II cabinets shall meet the grade B shock acceptance requirements of MIL-DTL-901.

4.5.1.1 Test plan. A shock test plan shall be prepared. Approval of the shock test plan from the Navy Preparing Activity shall be obtained prior to initiating shock testing. This test plan may be combined with the vibration test plan (see 4.5.2).

4.5.1.2 Test type. Shock testing shall be for class 1, type A equipment in accordance with MIL-DTL-901. For purposes of the shock test, the following MIL-DTL-901 classifications shall be applied to type I and type II cabinets:

a. Test category – medium weight

b. Mounting orientation – see table III

c. Mounting requirements – An adapter plate or baseplate shall be used in this test. The baseplate shall be in accordance with MIL-DTL-901. The baseplate shall be a square plate at least 1 inch thick. The test fixture baseplate shall be attached to the standard fixtures on the medium weight shock test machine. The cabinet shall be welded to the baseplate.



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TABLE III. Vertical axis specified test schedule.

| Test axis | Group I | Group II | Group III |
|---|---|---|---|
| Vertical | X | X | X |
| 30 degree 1/ | | X | X |
| 30 degree rotated 1/ | | X | X |

NOTE:
1/ With cabinet door facing downwards toward floor.

4.5.1.3 Loading of cabinet during test. During shock testing, the cabinet shall be loaded to its capacity with dummy weights. The dummy weights to be used in this testing are 1-gallon cans filled with water. The cans shall be filled using a graduated measuring cup or beaker so that the liquid level is as close to 1 gallon as possible. The water in the containers shall be dyed to make it more visible to observers and in photographs. The cans shall be evenly distributed on the shelves, but no cans shall be double stacked on the same shelf. The containers to be placed inside the cabinet during testing shall be a combination of new, unused metal paint cans with lids and metal jugs with caps. Each paint can shall have the lid secured using six lid clips per gallon can. After loading the cabinet, the door(s) of the cabinet shall be closed and latched, but not padlocked or secured by any other external means.

4.5.1.4 Monitoring during shock test. The cabinet under test shall be monitored visually after each hammer blow. Damage shall be noted by the test technician in writing in the test log and with photographs. Neither instrumentation nor transducers will be required during the testing. The cabinet door shall not be opened at any point during the seven test blows to inspect the dummy weights inside the cabinet.

4.5.1.5 Post-shock test observations.

a. At the completion of the test blows, the baseplate and the cabinet shall be removed from the test fixture and placed on a level surface. At this point, the cabinet door shall be opened and the contents of the cabinet shall be documented with photographs. The dummy weights shall be removed from the cabinet; use care when removing the dummy weights so as not to spill additional liquid. All the liquid remaining in the dummy weight cans shall be measured and recorded. In addition, any liquid in the cabinet sump shall be vacuumed out, and its volume measured and recorded.

b. When the cabinet is empty of the dummy weights and any remaining liquid has been vacuumed out, the interior of the cabinet shall be visually inspected. Damage shall be recorded in writing in the test log and with photographs.

c. Conduct a leak test of the cabinet sump. Fill the sump to its capacity with liquid and observe for at least 15 minutes to see if the sump is leaking.

4.5.2 Vibration test. Cabinets shall meet type I requirements of MIL-STD-167-1 for environmental vibration within a test range of 4 to 50 Hertz (Hz).

4.5.2.1 Test plan. A vibration test plan shall be prepared. Approval of the vibration test plan from the Navy Preparing Activity shall be obtained prior to initiating testing. This test plan may be combined with the shock test plan (see 4.5.1).

4.5.2.2 Test type. For purposes of the vibration test, the following shall be applied to type I and type II cabinets:

a. Mounting requirements – An adapter plate or baseplate shall be used in this test. The baseplate shall be a square plate at least 1 inch thick. The test fixture shall be attached to the platform of the vibration test machine. The cabinet shall be welded to the baseplate.

b. Equipment orientation – For these tests, the vibration axes shall be vertical, horizontal front-to-back, and horizontal side-to-side.

c. Conduct exploratory, variable frequency, and endurance testing.



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4.5.2.3 Loading of cabinet during test. During vibration testing, the cabinet shall be loaded to its capacity with dummy weights. The dummy weights to be used in this testing are 1-gallon cans filled with water. The cans shall be filled using a graduated measuring cup or beaker so that the liquid level is as close to 1 gallon as possible. The water in the containers shall be dyed to make it more visible to observers and in photographs. The cans shall be evenly distributed on the shelves, but cans shall not be double stacked on the same shelf. The containers to be placed inside the cabinet during testing shall be a combination of new, unused metal paint cans with lids and metal jugs with caps. Each paint can shall have the lid secured using six lid clips per gallon can. After loading the cabinet, the door of the locker shall be closed and latched, but not padlocked.

4.5.2.4 Monitoring during test. The cabinet under test shall be monitored visually and with accelerometers during the exploratory and variable frequency tests to determine if resonances exist.

4.5.2.5 Post-vibration test inspection.

a. At the completion of all testing in all axes, the cabinet door shall be opened and the contents of the locker shall be documented with photographs. The dummy weights shall be removed from the cabinet; use care when removing the dummy weights so as not to spill additional liquid. All the liquid remaining in the dummy weight cans shall be measured and recorded. In addition, any liquid in the cabinet sump shall be vacuumed out and its volume measured and recorded.

b. When the cabinet is empty of the dummy weights and any remaining liquid has been vacuumed out, the interior of the cabinet shall be visually inspected. Damage shall be recorded in writing in the test log and with photographs.

c. Conduct a leak test of the cabinet sump. Fill the sump to its capacity with liquid and observe for at least 15 minutes to see if the sump is leaking.

4.5.2.6 Acceptance criteria.

a. Mechanical soundness. The locker door shall not open, and the locker structure shall remain intact. All internal and external parts and accessories shall remain attached. The locker door shall be able to be opened at the conclusion of the testing.

b. Leakage. External spillage of contents shall not be permitted. Ruptured containers inside the locker shall not by themselves constitute a failure if the locker sump has retained the spill liquid.

c. Sump joints shall maintain their integrity and be capable of retaining liquids.

4.5.3 Spill containment. Evidence that the containment or sump shall hold a minimum of 10 percent of the aggregate cabinet volume for type I and type II cabinets shall be as specified (see 6.2). In addition, a sump leak test shall be conducted and test results documented. For the leak test, the cabinet shall be placed on a level surface. The sump shall be filled until it is on the verge of overflowing. The volume (in gallons) of water in the sump shall be recorded. With the locker sump filled to the verge of overflow, the locker shall stand for 15 minutes with no signs of external leakage.

4.5.4 Self-closing, self-latching doors. A functional test shall be conducted and test results documented (see 6.2) as evidence that cabinet doors can be locked.

4.5.5 Lockable. A functional test shall be conducted and test results documented (see 6.2) as evidence that cabinet doors can be locked.

4.5.6 Steel with acid-resistant epoxy paint. Evidence that cabinets are steel with acid-resistant epoxy paint shall be as specified (see 6.2).

4.5.7 Cabinet shelves. Evidence that cabinets are supplied with a minimum of two shelves shall be as specified (see 6.2). Documentation (see 6.2) shall also provide evidence that cabinet shelves are either galvanized metal or lined with acid-resistant epoxy paint or epoxy powder coating. The documentation shall include evidence that the cabinet shelves are fitted with polyethylene trays or liners to contain spillage. Direct observation of the cabinet shelves shall be allowed.

4.5.8 Color. Evidence that cabinets are in compliance with the color specifications (see 3.5.1) shall be as specified (see 6.2). In addition, direct observation of cabinet color shall be allowed.



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4.5.9 Identification plate(s)/label(s). Evidence that cabinets have identification plate(s)/label(s) (see 3.5.2) shall be as specified (see 6.2). In addition, direct observation of cabinet identification plate(s)/label(s) shall be allowed.

4.5.10 Hazard warning plate(s)/label(s). Evidence that cabinets have hazard warning plate(s)/label(s) (see 3.5.3) shall be as specified (see 6.2). In addition, direct observation of the cabinet hazard warning plate(s)/label(s) shall be allowed.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2).

6. NOTES

INFORMATION FOR GUIDANCE ONLY. (This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use.

a. Type I, class 1, class 2, and class 2 flammable cabinets are intended for storage of a limited supply of flammable and/or combustible material onboard ship. These cabinets are shock and vibration qualified. Limitations regarding shipboard installation locations can be found in Naval Ships' Technical Manual S9086-WK-STM-020, Afloat Hazardous Material Control and Management Guidelines Hazardous Materials Users Guide (HMUG).

b. Type II, class 1, class 2, and class 3 corrosive cabinets are intended for storage of a limited supply of corrosive material onboard ship. These cabinets are shock and vibration qualified. Limitations regarding shipboard installation locations can be found in S9086-WK-STM-020.

6.2 Acquisition requirements.

a. Title, number, and date of the specification.

b. Type and class required (see 1.2).

c. The specific issue of individual documents referenced (see 2.1 and 2.2).

d. Documentation (drawings, reports, etc.) to be submitted as evidence that requirements have been met (see 4.5.4 through 4.5.10).

e. Packaging requirements (see 5.1).

f. Requirements for installation instructions (see 6.5).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the supplier is called to this requirement, and manufacturers are urged to arrange to have the products they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, and information pertaining to qualification of products may be obtained from that activity.



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6.4 Part identification number (PIN). The following part or identification number procedure is for government purposes and does not constitute a requirement for the contractor.

| **F** | **AA** | **-** | **C** | **-** | **101** | **-** | **X** | **X** |
|---|---|---|---|---|---|---|---|---|
| Prefix for Federal Specification | Alphabetical designation | - | Alphabetical designation | - | Specification number | - | Type | Class |
| Examples: The PIN for a flammable HM cabinet with a maximum capacity of 12 gallons would be FAA-C-101-A1. The PIN for a corrosive HM cabinet with a maximum capacity of 15 gallons would be FAA-C-101-B2. | | | | | | | | |

6.4.1 Types. The cabinet types PIN codes are as follows:

| **Type** | **Characteristic** | **Color** | **PIN Code** |
|---|---|---|---|
| Type I | Flammable cabinet, shock and vibration qualified | Yellow | A |
| Type II | Corrosive cabinet, shock and vibration qualified | Dark blue | B |

6.4.2 Classes. The cabinet classes PIN codes are as follows:

| **Class** | **Characteristic** | **PIN Code** |
|---|---|---|
| Class 1 | 12 gallon | 1 |
| Class 2 | 15 gallon | 2 |
| Class 3 | 30 gallon | 3 |

6.5 Drawings and instructions. Drawings, sketches, and/or instructions should provide installation foundation, mounting, and applicable corrosion control details (see 6.2).

6.6 Subject term (key word) listing.

Acid
Combustible
Corrosive
Flammable
Locker
Shock
Vibration



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MILITARY INTERESTS

Custodians:
Army – MI
Navy – SH
Air Force – 03

Review Activities:
Army – GL, AV, QM1
Navy – CG, YD

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
Navy – SH
(Project 7125-2017-001)

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